

Nim Project Progress

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Goals

- Find winning strategies for Misere and Normal Play
- Make 'game' for Misere or Normal Play

Winning Strategies - Normal Play

- Recap : win by taking last object
- If nim sum is 0
 - Nothing
- If nim sum not 0
 - Finish each turn with nim sum 0
 - Find heap where xor of its size and total nim sum is less than original size
- If 2 heaps left
 - Symmetric removal

Winning Strategies - Misere

- Same as normal play mostly
 - Take nim sum and remove so can have total sum of 0
- Different when one heap or all heaps have 1 object left
 - Make sure the total number of piles is **even** when it's your opponent's turn. This forces your opponent to take the last stone and lose

Program

- Up until winning strategy
- Decided to change so the computer essentially plays against itself
 - Do all possible moves after the computer makes its move
 - Create total list of all games played out and all combinations